

ORF 245 – Midterm Exam

Spring 2024

- Duration of examination: 80 minutes.
- This exam is both open-note and open-book. You may reference class notes, the textbook, previous homework, and precept notes. You may also use a calculator.
- You may **not** use any other online or external materials.
- Collaboration is (of course) not permitted. You should not discuss the exam questions with others until grades are released and solutions have been posted.
- All of your answers (and supporting work) should be written directly in this booklet. Please be sure to clearly box your final answers to each (sub-)question.
- Unless stated otherwise, please ensure you show all important steps in deriving your final answer.
- Please write out and sign the following pledge on the first page of your solutions:
“I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this examination.”

1. **(One More Observation; 25 pts)**

Let $x_1, \dots, x_n, x_{n+1}$ denote some values of interest, and define the following descriptive statistics:

$$\bar{x}_n = \frac{1}{n} \sum_{i=1}^n x_i, \quad \sigma_n^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}_n)^2.$$

Notice that these two statistics depend *only* on the first n numbers in the sample, and not on the final observation x_{n+1} .

Now, let $\bar{x}_{n+1}$ and σ_{n+1}^2 denote the sample mean and sample variance when the final observation x_{n+1} is taken into account:

$$\bar{x}_{n+1} = \frac{1}{n+1} \sum_{i=1}^{n+1} x_i, \quad \sigma_{n+1}^2 = \frac{1}{n+1} \sum_{i=1}^{n+1} (x_i - \bar{x}_{n+1})^2.$$

(a) Express $\bar{x}_{n+1}$ in terms of only n , $\bar{x}_n$ and x_{n+1} .

(b) Show that $\sum_{i=1}^n (x_i - \bar{x}_n) = 0$.

(c) Show that:

$$(n+1)\sigma_{n+1}^2 = n\sigma_n^2 + \frac{n}{n+1}(x_{n+1} - \bar{x}_n)^2.$$

Q1 Answers:

2. (Discrete Random Variables; 25 pts)

Airlines often oversell tickets. Suppose that a plane with 100 seats, 104 passengers have tickets. Let X be the number of ticketed passengers who actually show up for the flight. Assume that the probability mass function of X , denoted $p(x)$, is given by the following:

x	96	97	98	99	100	101	102	103	104
$p(x)$	.15	.12	.14	.25	.17	.06	.05	.04	.02

If every passenger who does not get a seat needs to be compensated for \$300, whereas every ticket was sold for \$200. Let Z be the profit or loss of the airline due to overselling of 4 tickets, namely:

$$Z = \begin{cases} 4 * 200 - (X - 100) * 300 & \text{if } X > 100, \\ 800 & \text{otherwise.} \end{cases}$$

- (a) Compute $\mathbb{P}(X \leq 100)$.
- (b) What is the probability mass function of Z ?
- (c) What is the expected profit, $\mathbb{E}[Z]$, and the variance of profit, $\text{Var}(Z)$?
- (d) What is $\mathbb{E}[Z|X \leq 100]$ – the expected profit conditional on the airline **not** being over capacity?
(*Hint: what is the conditional distribution of Z given that $X \leq 100$?*)
- (e) What is $\mathbb{E}[Z|X > 100]$?

Q2 Answers:

3. (Standardizing a Normal / Gaussian Random Variable; 25 pts)

Recall that the cumulative density function of a Normal / Gaussian random variable with mean μ_X and variance σ_X^2 is given by:

$$F(x; \mu_X, \sigma_X^2) = \frac{1}{\sqrt{2\pi\sigma_X^2}} \int_{-\infty}^x e^{-\frac{(u-\mu_X)^2}{2\sigma_X^2}} du.$$

Let $X \sim N(\mu_X, \sigma_X^2)$ and define $Z = \frac{X-\mu_X}{\sigma_X}$.

- (a) For some real number $z \in \mathbb{R}$, express $\mathbb{P}(Z \leq z)$ in terms of the CDF of X .
- (b) Show that $\mathbb{P}(Z \leq z) = \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-v^2/2} dv$, i.e. the CDF of Z is that of a standard normal random variable.

Note: Here you may **not** use the property from class stating that $\frac{X-\mu_X}{\sigma_X}$ has a standard normal distribution.

(Hint: substitute and perform a change of variables to simplify the term in the integral.)

Q3 Answers:

4. (Joint Discrete Random Variables; 25 pts)

Consider the two discrete random variables X and Y with the following joint mass function.

		y		
		1	2	3
x	1	0	$\frac{1}{12}$	$\frac{1}{6}$
	2	$\frac{1}{12}$	$\frac{1}{3}$	$\frac{1}{12}$
	3	$\frac{1}{6}$	$\frac{1}{12}$	0

- Find the marginal probability mass functions of both X and Y .
- Are X and Y independent random variables? Justify your answer.
- Find $\text{Var}(X)$ and $\text{Var}(Y)$.
- Compute $\text{Cov}(X, Y)$.
- Find the probability mass function of the random variable $Z = X - Y$.
- What are $\mathbb{E}[Z]$ and $\text{Var}(Z)$?

Q4 Answers:

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